

Namhoon Kim

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Education

Georgia Institute of Technology

Aug 2024 - present

Ph.D. Student in Machine Learning

- **Advisor:** [Sara Fridovich-Keil](#)
- **Research Interest:** Computational Imaging, 3D Reconstruction, Uncertainty Quantification

Yonsei University

Mar 2019 - Aug 2024

B.S. in Electrical and Electronic Engineering

- **GPA:** 3.94/4.3, *Magna Cum Laude* (Top 3 % in Engineering School), **Major GPA:** 4.13/4.3
- **Senior Thesis** (Advisor: [Seung Ah Lee](#)) End-to-End, Physics-Informed Lensless Segmentation Network.

Publications

Conferences

- S. Jeong, **N. Kim**, and S. Fridovich-Keil, “Revisiting Shearlet Thresholding for 3D Denoising,” *NeurIPS* 2026 (under review).
- D. Bao, L. Wang, **N. Kim**, S. Fridovich-Keil, and J. Romberg, “Spectral Bias Often Emerges from Eigenspace Structure in Neural Network Training,” *NeurIPS* 2026 (under review).
- G. Jeong, S. Cho, M. Shon, **N. Kim**, K. Kim, W. Hwang, K. Seo, K. Kim, W. Kim, D. Ha, P. Venkatesan, K. Youn, R. Cherukuri, Y. Wang, S. Datta, A. Khan, and S. Yu “Physics-informed AI Accelerated Retention Analysis of Ferroelectric Vertical NAND: From Day-Scale TCAD to Second-Scale Surrogate Model,” *ICMC*, 2026.
- **N. Kim***, N. Moeini*, J. Romberg, and S. Fridovich-Keil, “3D Field of Junctions: A Noise-Robust, Training-Free Structural Prior for Volumetric Inverse Problems,” *ECCV*, 2026.
- **N. Kim** and S. Fridovich-Keil, “Grids Often Outperform Implicit Neural Representation at Compressing Dense Signals,” *NeurIPS*, 2025.
- **N. Kim** and S. Fridovich-Keil, “Towards Distribution-Shift Uncertainty Estimation for Inverse Problems with Generative Priors,” *IEEE CAMSAP* (invited paper), 2025.
- K. Lee, J. Bae, **N. Kim**, N. Beak, T. Kim, W. Park and S. Lee, “Differentiable Lensless Camera Design with Extended Field of View,” *Photonics Technology (at Optics and Photonics Congress)*, 2023.
- N. Baek, K. Lee, J. Bae, J. Park, T. Kim, D. Bae, **N. Kim**, W. Park and S. Lee, “Single-shot fabrication of light modulating phase mask for lensless camera using off-the-shelf components,” *Photonics Technology (at Optics and Photonics Congress)*, 2023.

Journals

- S. Lee, **N. Kim**, and S. Yoo, “Design-Space Analysis of Plug-and-Play Diffusion for PDE-Constrained Inverse Problems,” *TMLR*, 2026 (under review).
- **N. Kim**, A. Pananjady, A. Pourmorteza and S. Fridovich-Keil, “Perfusion Imaging and Single Material Reconstruction in Polychromatic Photon Counting CT,” *Medical Physics*, 2025 (under review).

Workshops

- **N. Kim***, N. Moeini*, J. Romberg, and S. Fridovich-Keil, “3D Field of Junctions: A Noise-Robust, Training-Free Structural Prior for Low-Dose CT,” *4th workshop on Vision-based Industrial InspectiON (VISION) (at CVPR)*, 2026.
- **N. Kim** and S. Fridovich-Keil, “Uncertainty Quantification for Inverse Problems with Generative Priors under Distribution Shift,” *Statistical Frontiers in LLMs and Foundation Models (at NeurIPS)*, 2024.

Selected Honors and Awards

Next-generation Science and Technology Leader NET, Korean Federation of Science and Technology Societies	2025
◦ Awarded \$2,000 research grant from KOFST supported interdisciplinary research projects. “Efficient Management of Data Centers for Extreme Environments using Digital Twin Technology”	
2nd prize, ML@GT Student Conference Best Poster Award:	Awarded \$200 Sep 2025
2nd prize, Mobility SW Hackathon, Hyundai Mobis	Feb 2023
◦ Developed an In-vehicle Noise Resistant Voice Recognition model with deep neural network.	
◦ Awarded \$4,000 for the prize and registered as an excellent talented human resource in Hyundai.	
3rd prize, Yonsei Dream Library: Metaverse Content Production Contest	Feb 2022
◦ Awarded \$800 for prize for developing a metaverse virtual university library using Lua and Roblox IDE.	
◦ Established a server-client communication enabling the construction of a competitive studying system based on the exchange of data between users.	

Mentoring

Sungwon Jeong, M.S. student at Georgia Institute of Technology	Fall 2025 - present
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Scholarships

Yonsei Eminence Scholarship, Yonsei University	Fall 2022 - Spring 2024
◦ Awarded for academic excellence: \$3000 (2/3 of Tuition) at Spring 2023, Fall 2023; \$2000 at Spring 2024; \$1000 at Fall 2022	

Teaching Experiences

Peer Tutor of course Digital Signal Processing (EEE4420), Yonsei University	Winter 2024
◦ Provided 16 hours of tutoring for 10 undergraduate students	
Residential College Program: Arduino Programming & Operation, Yonsei University	Spring 2023
◦ Appointed as a tutor and assisted 30 hours of lecture for 60 freshmen	
Social Engagement: Yon-In High School Students Mentoring, Yonsei University	Fall 2019
◦ Mentored high school students during the semester	
Peer Tutor of course Electricity & Magnetism(1) (EEE2030), Yonsei University	Summer 2022
◦ Provided 16 hours of tutoring for 40 undergraduate students	

Leadership and Military Service

Republic of Korea Defense Intelligence Command (Military Service)	Jul 2020 – Jan 2022
<i>Army Sergeant, honorable discharge</i>	
YAI, Artificial Intelligence Club Yonsei University — President	Dec 2023 – Jun 2024
Roboin, Robotics Club Yonsei University — Executive Staff	Mar 2023 – Aug 2023
Jamsin High School — Student Vice President	2014

Skills

Programming: Python, MATLAB, C/C++, C#, Lua, Verilog
Software and Tools: PyTorch, Linux, HTML/JS/CSS, Latex, Unity, CAD (Blender, Fusion 360, PTC Creo)
Experiment: Optical systems setup, Arduino, Raspberry Pi, 3D Printing (Form3+, Ender)
Soft Skills: Time management, Leadership, Teamwork